

Ramesh Vairagar Software Engineer (.NET & GIS Developer)
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PROFESSIONAL SUMMARY

Backend Engineer with over two (2+) years of experience specializing in the .NET ecosystem and GIS technology. I focus on building and scaling high-performance APIs that power large-scale enterprise systems, including AT&T's internal network and asset-tracking platforms. My core expertise lies in ArcGIS Pro development and deploying geospatial services to ArcGIS Server and Enterprise. I have a proven track record of managing complex data, including designing systems that handle 80,000+ daily users and processing massive datasets of over 68,000 records per request. Whether it's migrating legacy systems to .NET Core or integrating Oracle Spatial databases with cloud services like Azure, I enjoy solving the technical challenges that keep enterprise operations running smoothly.

SKILLS

- **GIS Platform & SDKs:** ArcGIS Pro, ArcGIS Server (Service Deployment & Configuration), ArcGIS Enterprise, and ArcPy for automation
- **Backend Development:** .NET Core 8, ASP.NET Web API, Microservices Architecture, REST/SOAP APIs, and Migration of legacy .NET Framework to .NET Core
- **Programming Languages:** C# .NET, Python (Scripting & ArcPy), SQL, and working knowledge of Java.
- **Data & Spatial Databases:** Oracle Spatial & DB, Basic Knowledge of PostgreSQL/PostGIS, Entity Framework, and Stored Procedures/Triggers
- **Cloud & Integration:** Microsoft Azure (Azure App service, Service Bus, Blob Storage, Key Vault), OAuth 2.0, System Interoperability, and Asynchronous Messaging.
- **Tools & Frontend:** Git/GitHub, Postman, Visual Studio, WPF/XAML for UI customization, and basic knowledge of React/TypeScript for Experience Builder.

PROFESSIONAL EXPERIENCE

Tech Mahindra Ltd, Pune (Software Engineer)

Nov 2023 - Present

Client: AT&T (American Telephone and Telegraph Company, USA) – .NET & GIS Developer.

Project 1: Waldo Portal (High-scale backend system supporting AT&T's internal network planning and asset inventory)

- I built and maintained high-availability .NET backend services that support over 80,000 daily users within AT&T's internal engineering ecosystem.
- I led the migration of legacy .NET Framework APIs to .NET Core, which significantly improved system speed, scalability, and long-term reliability.
- I designed a microservices-based architecture, allowing different teams to deploy their features independently and making the overall system easier to manage.
- I handled large-scale data updates and ingestion by creating automated background jobs in .NET using Windows Task Scheduler and Oracle DB.
- I ensured the system stayed secure by using Azure Key Vault for credentials and Azure Blob Storage for managing large files.
- I set up automated error alerts for our services and batch jobs, which helped the team catch and fix issues much faster.

Project 2: Zero Demand (GIS-Based Backend Platform for identifying business opportunities in underground cable infrastructure)

- Architected and developed scalable .NET Core REST APIs from the ground up to manage and expose critical geospatial data for underground cable infrastructure.
- I optimized our backend services to handle heavy loads of 68,000+ records per request, using smart pagination and Entity Framework to keep the system fast and stable.
- I configured and deployed GIS services on ArcGIS Server, making sure that map layers and analysis tools were high-performing and easily accessible for users.
- I used ArcPy to create custom geoprocessing workflows that automate the analysis of demand and infrastructure.

- Developed custom automation using Oracle database triggers and stored procedures to handle backend data processing and enforce complex business rules for cable tracking
- Integrated ArcGIS Pro into the development workflow to analyze demand and track infrastructure, bridging the gap between raw spatial data and actionable business insights
- Secured API access by implementing centralized authentication and Role-Based Access Control (RBAC) to protect sensitive infrastructure information.

EDUCATION

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| • Bachelor of Technology (B.Tech) in Computer Science and Engineering
Dr. Babasaheb Ambedkar Technological University, Lonere | Year: 2023 8.39 CGPA |
| • Higher Secondary Certificate (HSC – Class 12)
Maharashtra State Board of Secondary and Higher Secondary Education | Year: 2019 74.77% |
| • Secondary School Certificate (SSC – Class 10)
Maharashtra State Board of Secondary and Higher Secondary Education | Year: 2017 92.20% |

CERTIFICATIONS

- [Microsoft Azure Fundamentals \(AZ-900\) — Microsoft](#)
- Build ASP.NET Core Web API – Scratch to Finish (.NET 8) — Udemy
- AI White Belt Certification – Tech Mahindra

ACHIEVEMENTS & INTERESTS

- **Bravo Award**, Tech Mahindra — recognized for consistent delivery and quality of work.
- Active Member, JOSH Initiative, Tech Mahindra — contributed to organizing internal engagement and cultural events.

LANGUAGE

- English
- Hindi
- Marathi